SATMODCLASS / SIGNAL COMPARISON

Technical research report | satmod-research-pdf-v1

reportVersion: satmod-comparison-v1
id: d65667e5-95a6-402c-a4e0-7768f826fd18
createdAt: 2026-11-12T14:44:51.422809+00:00
modelId: 1d-cnn
modelVersion: supplied-cnn-v1
modelHash: e9705c98275e26d4a60210b2d563c4cc32ef94f4f0368974f1168245f91cf33d
processingMs: 47.11190902162343
sampleRates: {"first": 8000.0, "second": 8000.0}
ratesEqual: True
sampleClocksCompatible: True
normalizedAxes: False
lengthsEqual: True
overlapSamples: 128
sampleMse: 7.896804704716885e-05
powerDifferenceDb: -0.745101881674266
confidenceDifference: -0.1279953122138977

FIRST SELECTED RECORDING

captureId: 7f8fc310-4d6c-434b-8a66-621b73fdfaa1
start: 0
length: 128
position: {"captureBoundariesWithinWindow": [], "captureBoundaryNote": null, "centerFrequency": null, "fileSample": 0, "globalSampleIndex": 0, "globalSampleIndexSource": "Recording sample index; no independent hardware counter supplied", "seconds": 0.0, "sourceSample": 0, "timestamp": null, "windowCrossesCaptureSegment": false}
inputHash: 1fb21c61da6eeb8f2e712ad667c0a857eaea9258a379bb0a579c749f15b566ff
modelWindowHash: 1fb21c61da6eeb8f2e712ad667c0a857eaea9258a379bb0a579c749f15b566ff
inputRetention: Raw selected samples remain in the referenced managed recording; retain/export the recording for independent replay.
source: Generated report fixture, not real satellite RF
sessionId: synthetic-report-example
hash: 23e92ab698b4ec11542212c8820702e6a0a3bc71373846754efb1717cfe560e2
sourceHash: 8b855a0d25019326f9ce0a3446ec9630d98f765bf343b42b4835b0a2f826de0c
sampleRate: 8000.0
sampleRateKnown: True
sourceOffset: 0
datatype: parsed application format
scale: 1
license: Synthetic demonstration fixture

FIRST PREDICTION

status: available
reason: Unavailable
predictionSource: stored matching analysis
predictedModulation: QPSK
confidence: 0.5449240207672119
entropyBits: 1.3228542016218974
probabilities: {"16-QAM": 0.09109178930521011, "BPSK": 0.3639841377735138, "QPSK": 0.5449240207672119}
processingMs: 3.907649021130055

timingScope: Stored historical single-window inference, not a new comparison measurement.
preprocessing: {"amplitudePolicy": "Preserve samples; training z-score only", "mean": -6.560127076227218e-05, "std": 0.006070038769394159, "windowLength": 128, "windowStart": 0}

FIRST COMPUTED QUALITY

signalPower: 6.51925582297108e-05
powerDb: -41.85801976418816
rmsAmplitude: 0.008074190871518384
bandwidthHz: 7937.5
estimatedSnrDb: None
qualityNote: Noise and SNR cannot be uniquely inferred from an arbitrary capture without a reference or noise-only region. Power is in relative sample units (not calibrated watts).

SECOND SELECTED RECORDING

captureId: 7f8fc310-4d6c-434b-8a66-621b73fdfaa1
start: 128
length: 128
position: {"captureBoundariesWithinWindow": [], "captureBoundaryNote": null, "centerFrequency": null, "fileSample": 128, "globalSampleIndex": 128, "globalSampleIndexSource": "Recording sample index; no independent hardware counter supplied", "seconds": 0.016, "sourceSample": 128, "timestamp": null, "windowCrossesCaptureSegment": false}
inputHash: df99aafa329cd0e7fd5e94e3851da5832b3d62cb0aada3287b0fc6e6da60dc5d
modelWindowHash: df99aafa329cd0e7fd5e94e3851da5832b3d62cb0aada3287b0fc6e6da60dc5d
inputRetention: Raw selected samples remain in the referenced managed recording; retain/export the recording for independent replay.
source: Generated report fixture, not real satellite RF
sessionId: synthetic-report-example
hash: 23e92ab698b4ec11542212c8820702e6a0a3bc71373846754efb1717cfe560e2
sourceHash: 8b855a0d25019326f9ce0a3446ec9630d98f765bf343b42b4835b0a2f826de0c
sampleRate: 8000.0
sampleRateKnown: True
sourceOffset: 0
datatype: parsed application format
scale: 1
license: Synthetic demonstration fixture

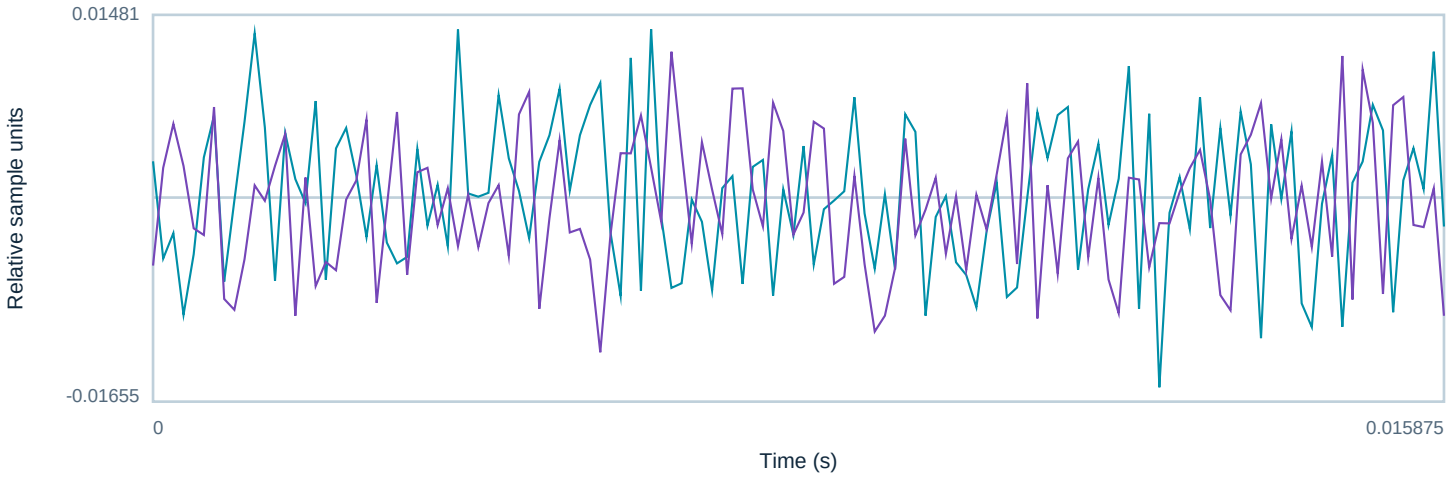
SECOND PREDICTION

status: available
reason: Unavailable
predictionSource: stored matching analysis
predictedModulation: BPSK
confidence: 0.6729193329811096
entropyBits: 1.192600592263891
probabilities: {"16-QAM": 0.09223425388336182, "BPSK": 0.6729193329811096, "QPSK": 0.23484642803668976}
processingMs: 3.512223018333316
timingScope: Stored historical single-window inference, not a new comparison measurement.
preprocessing: {"amplitudePolicy": "Preserve samples; training z-score only", "mean": -6.560127076227218e-05, "std": 0.006070038769394159, "windowLength": 128, "windowStart": 0}

SECOND COMPUTED QUALITY

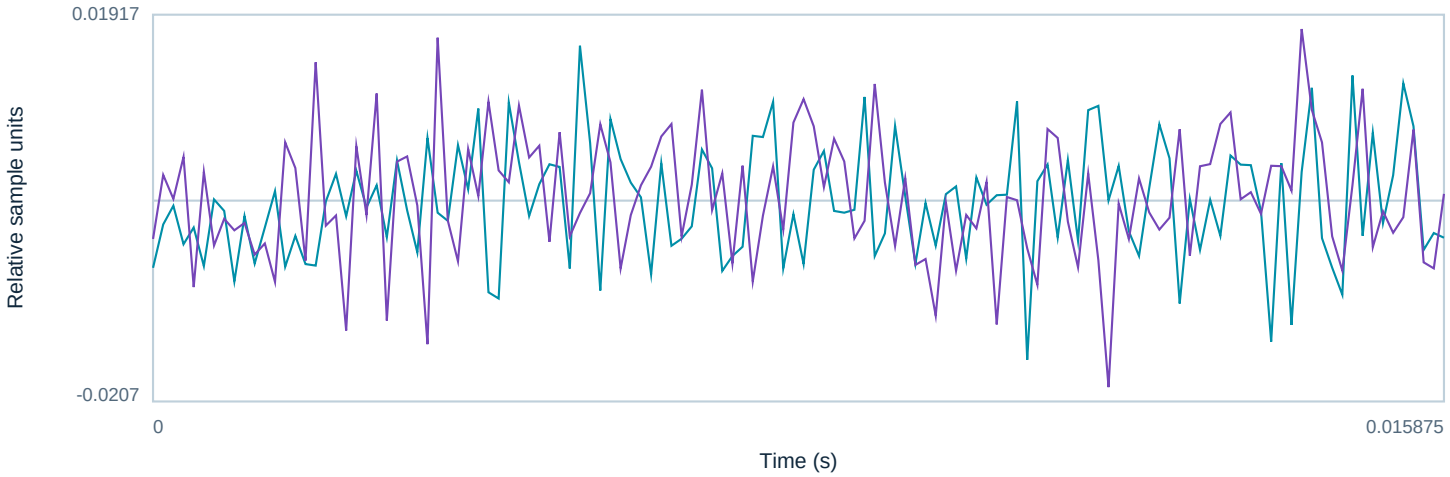
signalPower: 7.739416370216167e-05
powerDb: -41.11291788251389
rmsAmplitude: 0.008797395279408654
bandwidthHz: 7875.0
estimatedSnrDb: None
qualityNote: Noise and SNR cannot be uniquely inferred from an arbitrary capture without a reference or noise-only region. Power is in relative sample units (not calibrated watts).

FIRST I/Q WAVEFORM



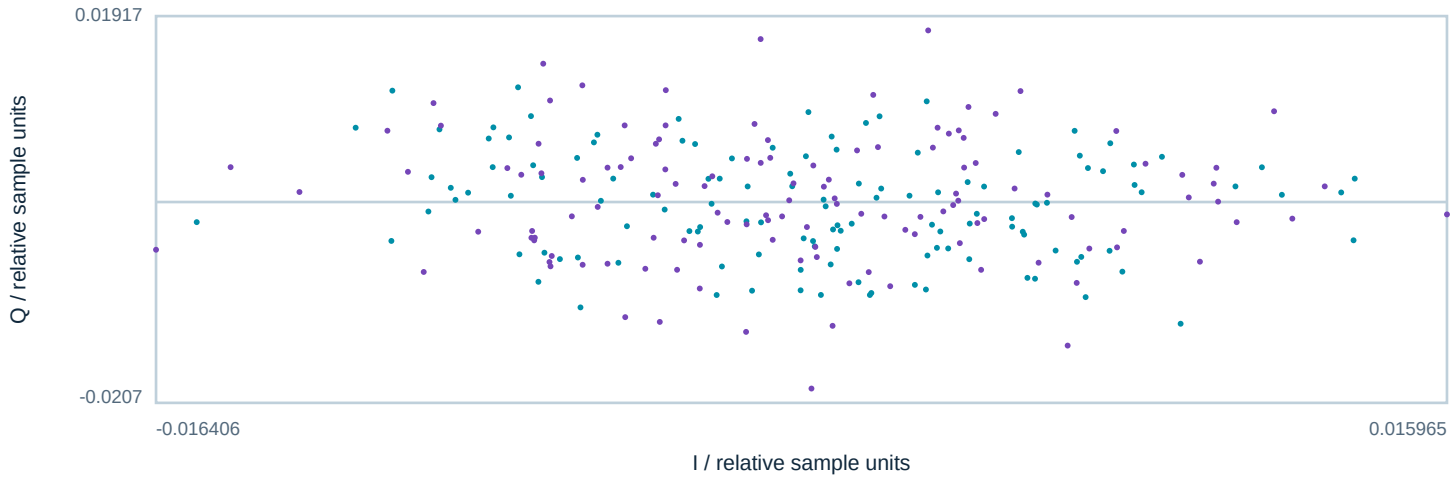
I | Q

SECOND I/Q WAVEFORM



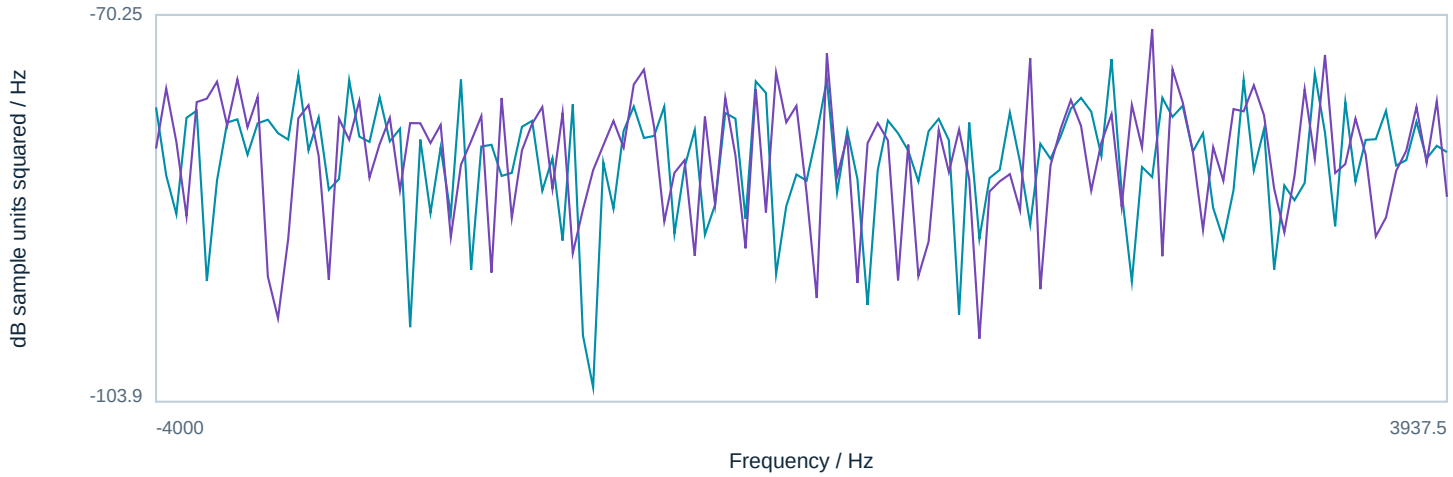
I | Q

RECEIVED CONSTELLATIONS



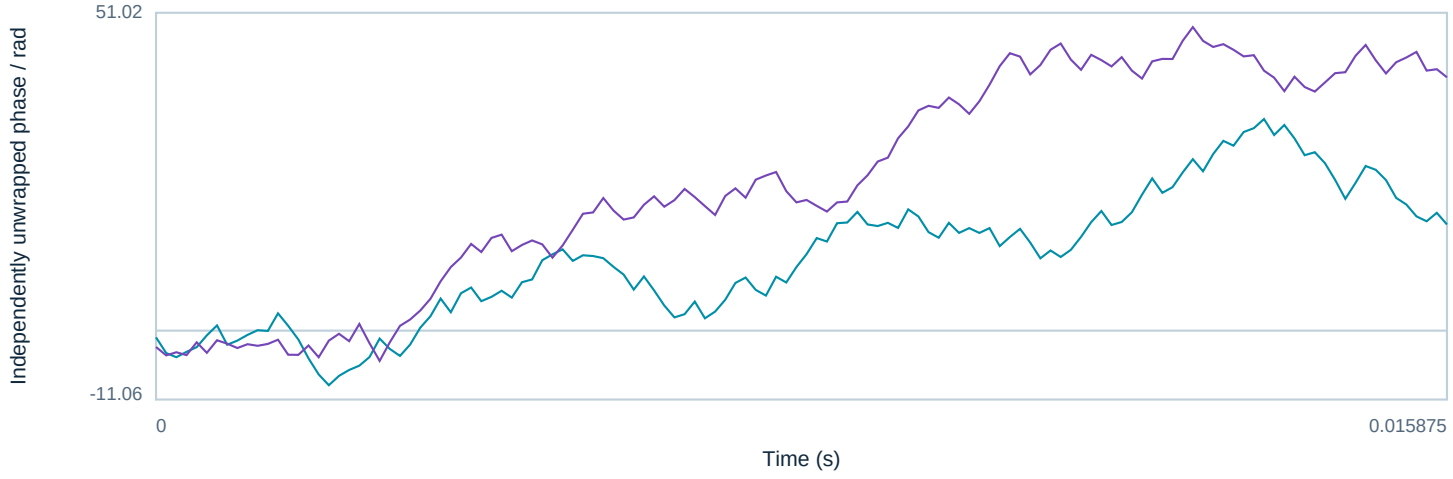
First | Second

SPECTRAL DENSITY COMPARISON



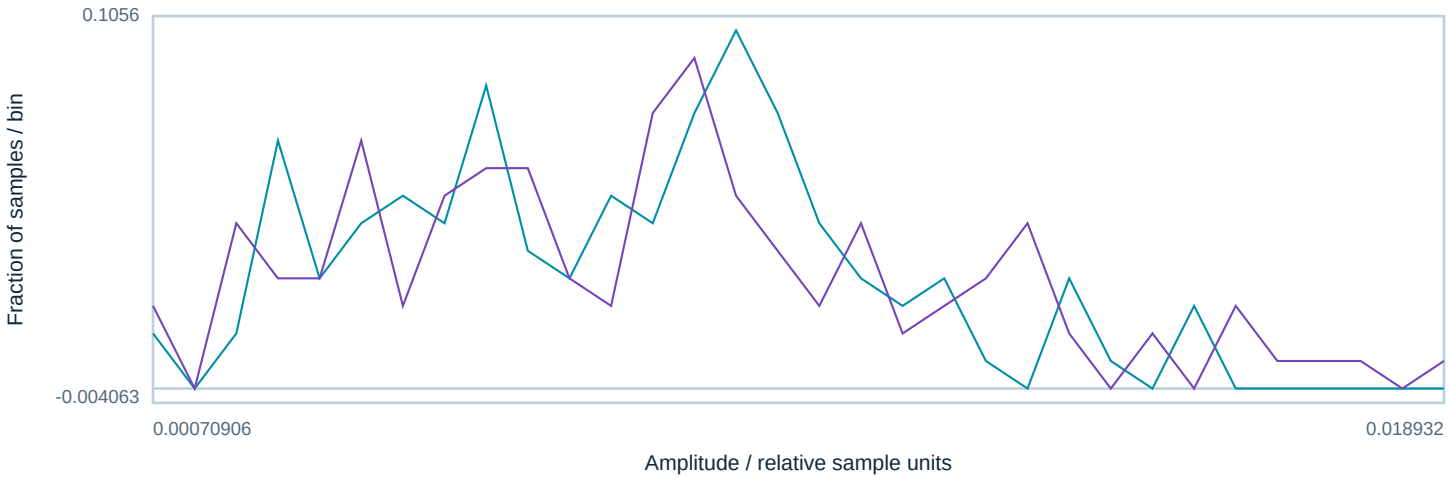
First | Second

PHASE TRAJECTORIES



First | Second

COMMON-BIN AMPLITUDE DISTRIBUTIONS



First | Second

Common amplitude bins; fractions normalize only bin counts, not input signal amplitudes.

ATTRIBUTION DIFFERENCES

status: available

reason: None

note: First minus second by relative model-window sample index only. No physical sample alignment, resampling or causal interpretation is inferred.

method: integrated-gradients

targetClass: QPSK

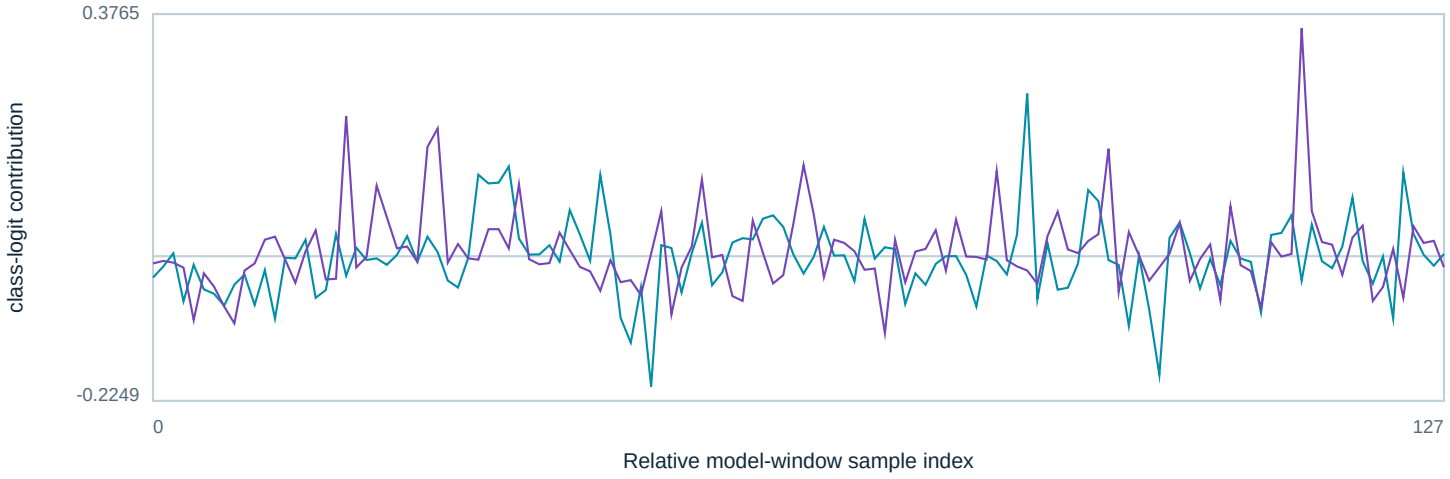
quantityUnit: class-logit contribution

firstAnalysisId: 1049c028-4cbd-454e-b0e8-07913f8f3149

secondAnalysisId: 93a9b69e-f493-4d16-b108-9beb2d1586ec

meanAbsoluteDifference: 0.04159932142175447

FIRST MINUS SECOND / integrated-gradients



I | Q

method: saliency

targetClass: QPSK

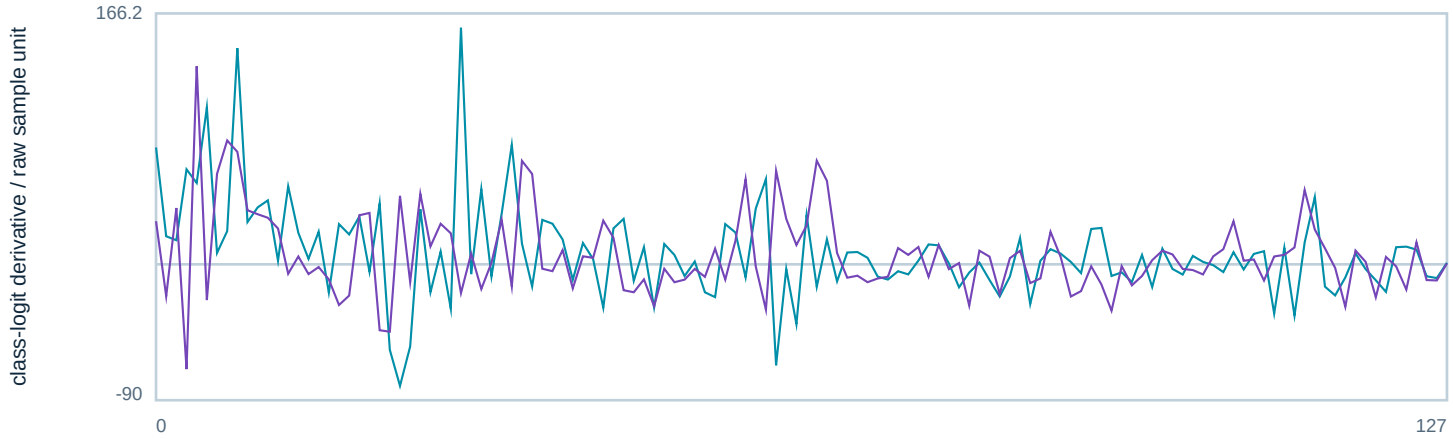
quantityUnit: class-logit derivative / raw sample unit

firstAnalysisId: 1049c028-4cbd-454e-b0e8-07913f8f3149

secondAnalysisId: 93a9b69e-f493-4d16-b108-9beb2d1586ec

meanAbsoluteDifference: 20.591467266443942

FIRST MINUS SECOND / saliency



I | Q

method: integrated-gradients

targetClass: BPSK

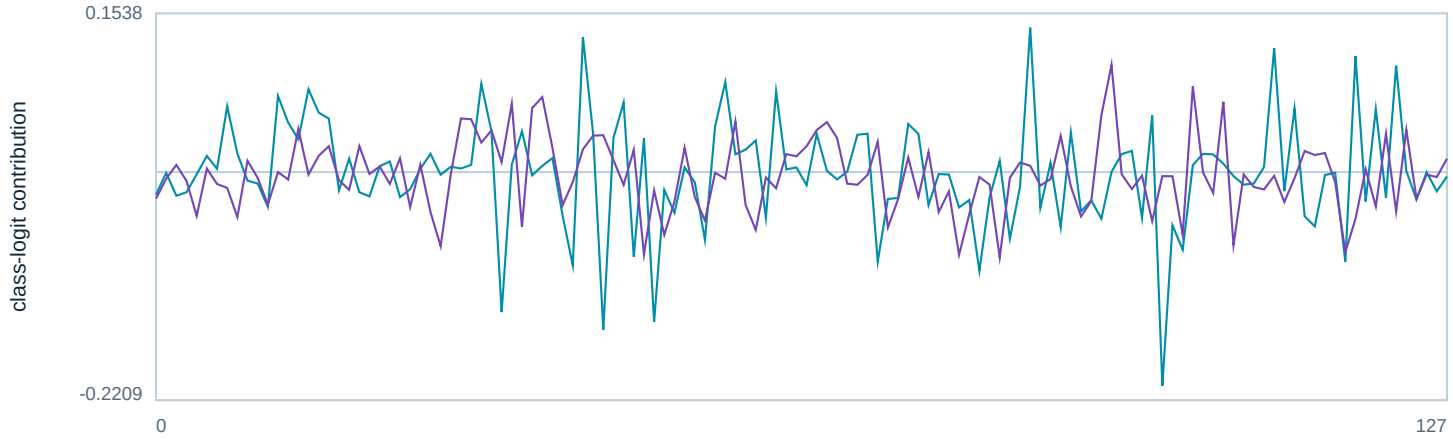
quantityUnit: class-logit contribution

firstAnalysisId: 1049c028-4cbd-454e-b0e8-07913f8f3149

secondAnalysisId: 93a9b69e-f493-4d16-b108-9beb2d1586ec

meanAbsoluteDifference: 0.031810102490762215

FIRST MINUS SECOND / integrated-gradients



I | Q

method: saliency

targetClass: BPSK

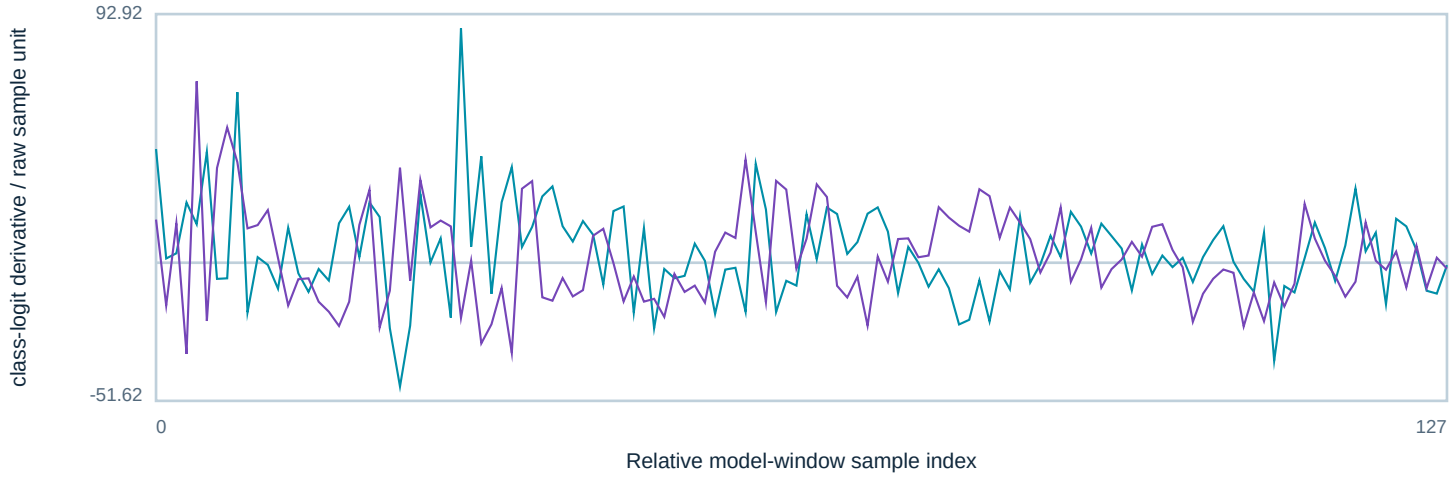
quantityUnit: class-logit derivative / raw sample unit

firstAnalysisId: 1049c028-4cbd-454e-b0e8-07913f8f3149

secondAnalysisId: 93a9b69e-f493-4d16-b108-9beb2d1586ec

meanAbsoluteDifference: 13.779923286320809

FIRST MINUS SECOND / saliency



I | Q

SCIENTIFIC LIMITATIONS

No sample alignment or resampling is inferred. MSE uses overlapping relative sample indices only and is not a modulation-similarity score.
Unequal or unknown clocks use normalized frequency/sample axes; spectral densities are transformed consistently.
All powers and amplitudes use relative recorder sample units, not calibrated RF watts.
Closed-set CNN outputs are uncalibrated; confidence differences do not establish a difference in correctness.
Phase is independently unwrapped in each window; near-zero samples have unavailable phase.
Plots draw at most 512 actual report samples per series. JSON retains the complete bounded plot arrays and exact input/capture/model identities.